CS/SE/CE 3354 Software Engineering

TemocTrade

Group Members:

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Shaikh Zaffin Raaid

Bilal Shah

Samara Sherven

1. TemocTrade

2. Delegation of tasks including each team member (first name and last name)

Keane Declan Ferdinand: Test Case and Implementation, editing the slides

Melanie A Lopez De Nava Reza: Method of estimates, cost of hardware products, use case diagram, slides creator, functional requirements

Peyton Caleb Mcentire: I'm working on the challenges, lessons learned, and conclusions

Shaikh Zaffin Raaid: I will be working on Comparison, Estimated cost of software products.

Bilal Shah: Compiling documents and committing to GitHub. Working on presentation slides: Objective, Non-functional requirements, Architectural Design.

Samara Sherven: I will be working on the project timeline, editing the project timeline slide, compiling the videos for the presentation, and editing the report.

3. [10 POINTS] Everything required and already submitted in Course Team Project Deliverable : link to previous deliverable

4. [35 POINTS – Broken down below] Project Scheduling, Cost, Effort, and Pricing Estimation, Project duration and staffing: Include a detailed study of project scheduling, cost, and pricing estimation for your project. Please include the following for scheduling and estimation studies:

a. [5 POINTS] Project Scheduling: Make an estimation of the schedule of your project. i. Please provide a start date, and end date by giving justifications about your estimation. ii. Also provide the details for: 1. Whether weekends will be counted in your schedule or not 2. What is the number of working hours per day for the project?

A simple project, with limited integration and basic features, will take 2 to 4 months [3]. The decided timeline is 14 weeks or 3 and a half months. The project started on February 8, 2026, when our team dedicated a week to requirement analysis and another 2 weeks to system design, after TemocTrade's proposal [3]. The team completed the majority of TemocTrade on the expected end date of April 5, 2026, in roughly 8 weeks [3]. This eight-week period was sufficient for backend and frontend development, and it corresponded with our estimated system completion. The last 3 weeks are split with 2 weeks dedicated to quality analysis and testing, and the last week to deployment and training [3]. Weekends are not counted in our schedule, since this project is discussed on weekdays, either face-to-face or online. The estimated number of working hours per day for this project is approximately 1-2 hours.

b.[15 POINTS] Cost, Effort and Pricing Estimation:

The method we have utilized to estimate the project cost is the Application Composition method. It is suitable as it is a smaller range project with a set amount of different states, components, and scenes. Additionally, the project is in its early stages and does not require very many complexities, which also means that there is room for reusable code implementation. First, you must collect and gather the exact planned amount of instances and other components of the project. This will include things like the home screen, the Student login, individual post screens, setting, checkout screen, transaction screen, customer-buyer chat, order confirmation, order view, etc. Given that we use, as students, relatively free infrastructure for our project, the only costs we are seeing would be our group opportunity costs, as well as cost forms $10-20 [1] on a domain for the application. Additionally, for image storage, we may have to include an additional $3-$5 per month. To calculate opportunity cost, we will set our value to around $15 an hour, and multiplied by 5 members, times the total hours spent on it in 14 weeks, totals to around $7,000. Assuming worst case scenario on the previously mentioned

prices, the total for the project would amount to $7,040. However, taking out opportunity costs, we would not be spending anymore than $2,000 to $3,000 realistically.

c.[5 POINTS] Estimated cost of hardware products (such as servers, etc.)

Our only hardware in this project that we already don't have, would be purchasing the domain for the application, in which for an annual cost, it would be around $12 annually.

d. [5 POINTS] Estimated cost of software products (such as licensed software, etc.)

The development of TemocTrade utilizes a combination of open-source tools, cloud services, and supporting software. While many tools are free, costs arise from hosting, database scaling, and productivity software.

Software and Cost Breakdown

- Development Tools: Visual Studio Code, Git, GitHub – Free
- Frameworks: React, Node.js, Express.js – Free
- Database: MongoDB – ~$50–60/month → ~$700/year
- Cloud Hosting: Amazon Web Services – ~$50/month → ~$600/year
- Payment System: Stripe – No fixed cost (transaction-based)
- Testing: Jest – Free
- Design Tools: Figma – ~$12/month → ~$144/year

Productivity Tools:

- Microsoft Office – ~$70/year
- Adobe Creative Cloud – ~$720/year
- ChatGPT – ~$240/year

Total Estimated Cost

Approximately $2,400–$2,500 per year

Conclusion

Most development tools are free; however, costs arise from hosting, database services, and optional productivity tools. The overall cost remains moderate and scalable for a university-based system.

e. [5 POINTS] Estimated cost of personnel (number of people to code the final product, training cost after installation)

If we hired a group of developers, we would assume that a relatively small group of developers, around 5 members, could make a well functioning MVP within a short time frame, especially using AI-assisted code development, and already having requirements and diagrams laid out for them. We estimate paying them $15 an hour, and multiplying by 5 members, times the total hours spent on it in 14 weeks, for roughly 7 hrs a week, totals to around $7,000.

5. A test plan for your software:

a. Describe the test plan for testing a minimum of one unit of your software.

The test plan for testing a module called validate_listing_price. The objective of this test plan is to verify the integrity of the model logic within our architecture. We focus on the product module and specifically the part about user-inputted prices. The test strategy is Unit Testing in Python with the unittest framework.[2] The unit we will test is the validate_listing_price. The function should return true for an int or float greater than 0 and false for anything that is not an int or float, or 0 or less. We shouldn't allow users to have negative pricing for products, or input strings, or any other data type.

b. [5 POINTS] As evidence, write a code for one unit (a method for example) of your software in a programming language of your choice, then use an automated testing tool (such as JUnit for a Java unit) to test your unit and present results.

unit:

```
def validate_lisiting(price):
    # this checks that price is an int or float and is greater than 0
    if isinstance(price, (int, float)) and price >0:
        return True
    else:
        return False
```

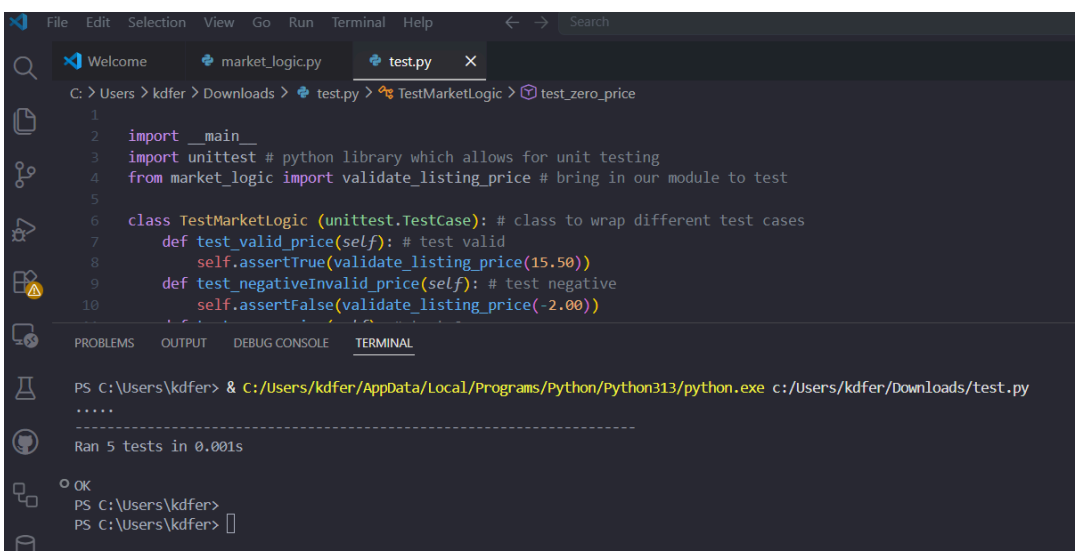
test script: **This code is also in test.py in github repo**

```
import __main__
import unittest # python library which allows for unit testing
from market_logic import validate_listing_price # bring in our module to test
```

```
class TestMarketLogic (unittest.TestCase): #  wrap different test cases
        def test_valid_price(self): # test valid
                self.assertTrue(validate_listing_price(15.50))
        def test_negativeInvalid_price(self): # test negative
                self.assertFalse(validate_listing_price(-2.00))
        def test_zero_price(self): # test 0
                self.assertFalse(validate_listing_price(0))
        def test_string_input(self): # test string input
                self.assertFalse(validate_listing_price("twenty dollars"))
        def test_none_price(self): # test none
                self.assertFalse(validate_listing_price(None))


# means you only run when script is run directly
if __name__ == '__main__':
        unittest.main()
```

When running the tests, here are the results:



OK signifies that all test cases passed.

c. [5 POINTS] Clearly define what test case(s) are provided for testing purposes and what results are obtained. (Ch 8)

The code for the test cases are above but we have a test case for the following:

- Valid entry.
- Invalid negative entry.
- Invalid 0 entry.
- String invalid entry.
- Empty (None) entry.

The results for our test show all test cases run as intended.

6. [5 POINTS] Comparison of your work with similar designs: a. This step requires a thorough search in the field of your project domain. Please cite any references you make.

To evaluate the effectiveness of TemocTrade, it is compared with existing marketplace platforms such as Craigslist, Facebook Marketplace, and OfferUp.

a. Craigslist

Craigslist is one of the earliest online classified advertisement platforms, allowing users to post and browse listings for goods and services.

Similarities with TemocTrade:

- Users can create and browse listings.
- Direct communication between buyers and sellers.
- Listings are user-generated.

Differences:

- No strong user authentication vs. TemocTrade's UTD email verification.
- Limited moderation vs. TemocTrade's reporting and safety features.
- Outdated interface vs. TemocTrade's modern and user-friendly design.

Conclusion:

TemocTrade improves upon Craigslist by providing greater security, usability, and a trusted user base.

b. Facebook Marketplace

Facebook Marketplace enables users to buy and sell items within local communities using their Facebook accounts.

Similarities with TemocTrade:

- Requires user authentication.
- Supports messaging between users.
- Listings include images and descriptions.

Differences:

- Broad audience vs. TemocTrade's student-only access.
- Social profile-based vs. institutional (UTD email) authentication.
- Large and cluttered listings vs. a focused marketplace.

Conclusion:
TemocTrade offers a more secure and relevant environment tailored to university students.

c. OfferUp
OfferUp is a mobile-first marketplace focused on local buying and selling through a modern interface.

Similarities with TemocTrade:

- Local peer-to-peer transactions.
- Messaging between buyers and sellers.
- Mobile-friendly design.

Differences:

- Open to all users vs. TemocTrade's restricted student access.
- Optional verification vs. enforced authentication.
- General marketplace vs. academic-focused environment.

Conclusion:
TemocTrade provides a more controlled and community-focused platform with improved trust.

Summary
Overall, while existing platforms provide general marketplace functionality, TemocTrade differentiates itself through verified users, increased security, and a student-focused experience.

7. [5 POINTS] Conclusion: a. Please make an evaluation of your work. i. Describe any changes that you needed to make (if any), if things have deviated from what you had originally planned. ii. Give justification for such changes.

TemocTrade developed from an initial project concept into a more complete and realistic software engineering design throughout Deliverable 1 and Deliverable 2. In Deliverable 1, the team focused on establishing the project's foundation by defining the system scope, assumptions, software requirements, UML diagrams, and MVC architecture. In Deliverable 2, the project was expanded by adding more practical and implementation-focused elements such as project scheduling, cost and effort estimation, hardware and software cost analysis, personnel cost planning, and a test plan with unit testing. The project did not significantly change, since the original concept and feedback remained consistent, but the project became much more detailed, structured, and realistic as new planning and testing components were added. These changes helped transform TemocTrade from a conceptual design into a stronger and more feasible proposal for a UTD student marketplace system.

8. [5 POINTS] References: a. Please include properly cited references in IEEE paper referencing format. (You may see a referencing example in the sample IEEE paper in URL:

[1] S. MacDonald, "How much does a domain name cost?," Squarespace, https://www.squarespace.com/blog/how-much-does-domain-name-cost#:~:text=The%20average%20cost%20of%20a,Newer%20extensions%2C%20such%20as%20. (accessed Apr. 5, 2026).

[2] Python Software Foundation, "unittest — Unit testing framework," *Python Documentation*. [Online]. Available: https://docs.python.org/3/library/unittest.html. [Accessed: Apr. 5, 2026].

[3] Keyhole Software, "Custom Software Development Cost: 2026 Pricing & Timeline Benchmarks," *Keyhole Software*, Jan. 14, 2026. https://keyholesoftware.com/cost-custom-software-development/. Accessed Apr. 5, 2026.